Railway 110VDC to 24V 100kHz Output Converter

Project Handover and Remaining Engineering Work

Prepared for: External engineering / PCB design partner
Prepared by: Electro Repairs
Purpose: Continue and finalize the railway power converter redesign
Updated requirement: Output must change from 12V 100kHz to 24V 100kHz

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Executive Summary

The current prototype confirmed that the architecture can generate a high-frequency square-wave output, but it also exposed several required design changes before the design can be considered production-ready.

The most important findings are:

1. **The LTC7897 buck stage requires proper gate-drive tuning.**
2. **The low-side buck MOSFET Q2 requires mandatory gate damping.**
3. **The oscillator output resistor must be corrected.**
4. **The LTC7897 driver supply voltage must be tunable.**
5. **The layout around the LTC7897, buck MOSFETs, sense resistor, and switch node requires improvement.**
6. **The design must now be revised for a 24V output instead of the original 12V output.**

The existing prototype has been heavily reworked and should be treated as a proof-of-concept and fault-finding board, not as a final validation platform.

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1. Project Overview

This project is a redesign of an obsolete railway electronics PCB. The design accepts a nominal 110VDC input and must generate a high-frequency square-wave output.

The current prototype architecture is:

110VDC input

→ LTC7897 synchronous buck converter

→ DC bus rail

→ LM5106 half-bridge driver

→ MOSFET half-bridge output

→ square-wave output

The present prototype was originally developed for a 12V 100kHz square-wave output, but the project requirement has now changed to:

Required final output: 24V, 100kHz square wave

Unless otherwise specified by Electro Repairs, this should be treated as a 0V to 24V square wave referenced to output ground, not an isolated output and not a ±24V bipolar waveform. This must be confirmed before final design release.

The previous 12V design cannot simply be scaled by changing the feedback divider. The buck converter, half-bridge driver supply, MOSFET voltage ratings, TVS clamp, capacitors, current limit, and thermal design must all be reviewed for 24V operation.

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2. Current Prototype Status

The current prototype successfully demonstrated the main architecture. After hardware modifications, the board produced the expected 12V 100kHz output waveform.

However, the prototype also exposed several design issues that require a proper PCB revision. The current PCB has been heavily reworked around the LTC7897 controller and should no longer be treated as a clean validation platform.

The provided project files show:

- **U1: LTC7897RUFD#TRPBF buck controller**
- **Q1: BSC110N15NS5 high-side buck MOSFET**
- **Q2: ISC055N15NM6ATMA1 low-side buck MOSFET**
- **U8: LM5106SDX/NOPB half-bridge driver**
- **Q5/Q6: BUK7Y6R0-60EX output half-bridge MOSFETs**
- **Rsense: 8mΩ buck current-sense resistor**
- **R30: 20mΩ output current-sense resistor**

- **D3: SMBJ18CA output TVS**
- **R20: 220kΩ series resistor in the oscillator output path in the original schematic**
- **C14/R11: SW-node snubber network, 470pF + 10Ω**

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3. Prototype Rework Summary

Several temporary reworks were carried out during prototype debugging. These changes were made directly on the assembled PCB and should be included properly in the next schematic and PCB revision.

The most important successful reworks were:

Q2 low-side buck MOSFET gate damping added:

BGUP = 10Ω

BGDN = 2Ω

Q2 gate-source pulldown = 10kΩ

Q1 high-side buck MOSFET gate drive modified:

TGUP = 2Ω

TGDN = 2Ω

Oscillator output resistor corrected:

f_OUT series resistor changed from 220kΩ to 100Ω

SW-node snubber capacitor removed:

C14 removed / DNP

Half-bridge output stage enabled and verified:

12V 100kHz output waveform produced correctly

Additional experiments were performed on the LTC7897 driver-supply configuration:

DRVSET / DRVUV were tested in different configurations.

Lower DRVCC reduced LTC7897 heating but increased Q2 heating.

Higher DRVCC improved Q2 drive but increased LTC7897 heating.

This indicates that the next design must include proper DRVSET / DRVUV configuration options so the final driver voltage can be selected based on measured temperature, waveform quality, and efficiency.

During prototype debugging, multiple LTC7897 devices were exposed to abnormal operating conditions and fine-pitch rework. Any failures on the heavily reworked boards should not be treated as clean production-validation failures; they should be treated as prototype stress and rework events.

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4. Important Findings From Prototype Testing

4.1 Buck low-side MOSFET gate drive issue

The original PCB did not include a proper series gate resistor for Q2, the low-side buck MOSFET. This caused excessive switching stress and rapid Q2 heating.

A temporary hardware modification was made:

BGUP = 10Ω

BGDN = 2Ω

Q2 gate-source pulldown = 10kΩ

After adding gate damping to Q2, the immediate Q2 overheating issue was greatly improved and the converter could produce the 100kHz output.

This change must be included properly in the next PCB revision.

4.2 High-side buck MOSFET gate drive should also be revised

The original design had a small 2Ω resistor in the Q1 gate path, but the next revision should use the LTC7897 split gate-driver pins properly.

Recommended structure:

TGUP → turn-on resistor → Q1 gate

TGDN → turn-off resistor → Q1 gate

Q1 gate → 10kΩ → Q1 source / SW

4.3 Oscillator output series resistor was incorrect

The original schematic shows:

R20 = 220kΩ

in the f_OUT signal path. This is too high for driving the LM5106 input and can make the logic signal noise-sensitive or cause missed pulses.

The prototype was reworked to:

R20 = 100Ω

Recommended final value:

R20 = 100Ω to 330Ω

Recommended: 100Ω or 220Ω

4.4 LTC7897 heating requires design attention

The LTC7897 became the main heating component after the Q2 gate-drive problem was improved. This must be addressed through:

- **Correct gate-drive resistor networks**
- **DRVCC voltage optimization**
- **Improved LTC7897 thermal layout**
- **Proper exposed-pad soldering and thermal vias**
- **Input-current and temperature validation under real load**

The prototype showed a trade-off between LTC7897 heating and Q2 heating:

Lower DRVCC:

LTC7897 runs cooler

Q2 may heat more

Higher DRVCC:

Q2 runs cooler

LTC7897 may heat more

The final design must include selectable DRVSET / DRVUV configuration options so the driver voltage can be tuned correctly.

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5. Required Change: Output Must Become 24V 100kHz

The new output requirement is:

24V square wave

100kHz switching frequency

This is not a simple connector change. The whole power path must be reviewed.

The external engineering partner must confirm whether the final output is:

0V to 24V square wave

or another waveform definition.

The required output current must also be confirmed before final component selection. The previous prototype was evaluated around the original 12V output requirement, but the 24V output power requirement may be significantly higher depending on the load current.

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6. Required Electrical Changes for 24V Output

6.1 Buck converter output voltage change

The LTC7897 buck stage must be changed from a 12V bus to a 24V bus.

The present feedback divider uses:

R29 = 140kΩ

R4 = 10kΩ

which sets the buck output around 12V.

For 24V output, using the normal feedback equation:

VOUT = 0.8V × (1 + RTOP / RBOTTOM)

If:

RBOTTOM = 10kΩ

then the required top resistor is approximately:

RTOP = 290kΩ

Recommended starting value:

Rbottom = 10kΩ

Rtop = 287kΩ or 294kΩ, 1%

The exact value should be selected based on the required 24V tolerance.

6.2 Inductor and current-limit recalculation

The present buck inductor is:

L1 = Würth 74435584700

and the buck current sense resistor is:

Rsense = 8mΩ

Changing from 12V to 24V roughly doubles the output power for the same output current. The inductor ripple current, current-limit margin, saturation current, RMS current, and thermal rise must all be recalculated.

The current 8mΩ sense resistor may no longer provide enough peak-current margin for a 24V output, especially if the final output must support several amps.

The external engineering partner must recalculate:

inductor ripple current

inductor saturation current

buck peak current limit

MOSFET RMS current

sense resistor dissipation

thermal margin

6.3 EXTVCC at 24V

The LTC7897 EXTVCC pin may be connected to the 24V bus only if all datasheet limits and transients are respected.

A 24V rail is close enough to the EXTVCC operating limit that transients, ringing, and overshoot must be checked carefully.

The final design should protect EXTVCC from excessive transient voltage.

6.4 Half-bridge driver supply must not become 24V

This is critical.

The LM5106 VDD supply must not be tied directly to the new 24V bus.

For the 24V output version, the output half-bridge power rail may be 24V, but the LM5106 driver supply must remain around:

10V to 12V

Required change:

24V_BUS → separate 10V/12V driver supply → LM5106 VDD

The high-side MOSFET drain can connect to 24V_BUS, but the LM5106 VDD pin must use a regulated gate-driver supply.

Any circuit currently powered from the 12V bus must be reviewed before changing the bus to 24V. This includes the LM5106 supply, 5V regulator input, oscillator supply path, comparator supply path, bootstrap components, pullups, clamps, and decoupling components connected to the bus.

6.5 Output MOSFET voltage rating review

The current output half-bridge MOSFETs are:

Q5/Q6 = BUK7Y6R0-60EX

These are 60V-class MOSFETs. For a 24V square-wave output, they may still work electrically, but ringing, cable transients, and TVS clamp voltage must be reviewed carefully.

Recommendation:

Review whether 60V MOSFETs provide enough margin.

Consider 80V or 100V MOSFETs if cable/load transients are significant.

6.6 Output TVS must change

The current output TVS is:

D3 = SMBJ18CA

This is not suitable for a 24V square-wave output because it may conduct during normal operation.

For a 24V output, select a TVS with a reverse standoff voltage above the normal output voltage.

Possible starting point:

SMBJ30CA

SMBJ33CA

The final value must be selected based on allowable output overshoot and MOSFET voltage rating.

6.7 Output capacitors must be voltage-derated

The existing design has multiple 10µF and 100µF output/bus capacitors.

For a 24V bus, all bus capacitors must be reviewed for:

voltage rating

DC bias derating

ripple current

temperature rating

lifetime

Recommendation:

Use at least 50V rated capacitors on the 24V bus.

Consider 63V parts if space and cost allow.

6.8 5V regulator input rating and dissipation

The design uses:

U7 = TLV76050DBZR

as the 5V regulator.

At 24V input, the LDO dissipates:

P = (24V - 5V) × Iload

If the 5V load is more than a few milliamps, the LDO may get hot. A small buck regulator may be preferable for the final design.

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7. Required Buck Gate-Drive Revision

The next PCB must include proper gate-drive resistor networks.

7.1 Recommended Q1 network

TGUP → 4.7Ω → Q1 gate

TGDN → 2.2Ω → Q1 gate

Q1 gate → 10kΩ → Q1 source / SW

Allow tuning options:

TGUP: 2Ω, 4.7Ω, 10Ω

TGDN: 0Ω, 1Ω, 2.2Ω

7.2 Recommended Q2 network

BGUP → 10Ω → Q2 gate

BGDN → 2.2Ω → Q2 gate

Q2 gate → 10kΩ → Q2 source / GND

Allow tuning options:

BGUP: 10Ω, 15Ω, 22Ω

BGDN: 0Ω, 1Ω, 2.2Ω

Q2 gate damping is mandatory. The prototype showed that no Q2 gate damping caused severe heating and instability.

The gate-source pulldown resistors should connect directly between gate and source of each MOSFET:

Q1 gate-source pulldown: Q1 gate → Q1 source / SW

Q2 gate-source pulldown: Q2 gate → Q2 source / GND

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8. Required LTC7897 Driver-Supply Options

The next PCB must include configuration options for:

DRVSET

DRVUV

Recommended options:

DRVSET:

- GND

- resistor to GND for intermediate DRVCC

- INTVCC

DRVUV:

- GND

- floating

- INTVCC

The prototype showed a trade-off:

Low DRVCC:

LTC7897 runs cooler

Q2 may heat more

High DRVCC:

Q2 runs cooler

LTC7897 may heat more

Recommended intermediate test target:

DRVCC ≈ 8V to 9V

The final DRVCC value must be chosen based on measured LTC7897 temperature, Q1/Q2 temperature, switching waveform quality, and efficiency under representative load.

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9. Current-Limit and EN Control Improvements

The design includes a TLV3201 comparator and a 20mΩ output shunt for output-stage current protection.

The current circuit should be improved for noise immunity.

Recommended additions:

1. Add hysteresis resistor footprint:

TLV3201 output → OP+

2. Make C17 configurable:

1nF / 4.7nF / 10nF

3. Make PGOOD-to-EN clamp isolatable:

add 0Ω / DNP option

4. Add EN test point close to LM5106

Initial recommended values:

R31 = 1kΩ

C17 = 4.7nF

Rhyst = 1MΩ

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10. Snubber and Clamp Strategy

The current SW snubber is:

R11 = 10Ω

C14 = 470pF

During prototype testing, removing C14 reduced input current. Therefore, C14 should not be fitted by default until SW-node ringing is measured.

Recommended next revision:

R11 footprint fitted

C14 DNP by default

Optional C14 values: 100pF, 220pF, 470pF

The output snubber and TVS must be redesigned for the new 24V output.

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11. Thermal Layout Requirements

The next revision must improve thermal layout around:

LTC7897

Q1

Q2

L1

Rsense

Q5/Q6

LM5106

For U1, the LTC7897:

- Use large exposed-pad copper area.

- Use multiple thermal vias under and around the exposed pad.

- Connect to a solid bottom-side copper plane.

- Avoid unnecessary thermal relief on high-current/thermal pads.

- Keep DRVCC, INTVCC, and EXTVCC capacitors close to U1.

The current prototype required repeated rework on QFN pins, so the current board is no longer a reliable thermal validation platform.

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12. Test Points Required in Next Revision

The current PCB already includes several useful test points such as:

TP_12V_BUS

TP_DRVCC

TP_INTVCC

TP_PGOOD

TP_RUN

TP_OVLO

TP_VFB

TP_ITH

The next revision should add or preserve:

TP_VIN_110V

TP_GND

TP_24V_BUS

TP_EXTVCC

TP_DRVCC

TP_INTVCC

TP_SW

TP_Q1_GATE

TP_Q2_GATE

TP_PGOOD

TP_RUN

TP_OVLO

TP_VFB

TP_ITH

TP_EN

TP_f_OUT

TP_OUT_24V_SQ

TP_OUT_GND

TP_OP+

TP_OP-

TP_COMP_OUT

These are required for safe debugging and validation.

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13. Recommended Development Steps Remaining

Step 1: Confirm final requirements

The external engineering partner must confirm:

Input voltage range

Input transient requirements

Output waveform definition

Output voltage tolerance

Output current requirement

Duty cycle requirement

Load type

Cable length

Protection requirements

Temperature range

Mechanical size limits

Connector requirements

The most important open item is whether the output is:

0V to 24V square wave

or something else.

The required output current must be confirmed before final component selection. The previous prototype was evaluated around the original 12V output requirement, but the 24V output power requirement may be significantly higher depending on the load current.

Step 2: Redesign schematic for 24V output

Required schematic changes:

1. Change buck feedback divider for 24V_BUS.

2. Recalculate inductor and current sense for 24V output power.

3. Add proper Q1/Q2 split gate-drive resistor networks.

4. Add Q1/Q2 gate-source pulldowns.

5. Correct f_OUT series resistor to 100Ω–330Ω.

6. Add DRVSET/DRVUV configuration options.

7. Add separate 10V/12V supply for LM5106 VDD.

8. Replace output TVS for 24V operation.

9. Recheck all output capacitors for 24V bus rating.

10. Recheck 5V regulator dissipation from 24V bus.

11. Improve comparator filtering and hysteresis.

12. Add isolation options for PGOOD-to-EN and comparator-to-EN paths.

Step 3: Re-layout PCB

PCB layout must follow high-voltage synchronous buck layout rules.

Critical layout areas:

1. VIN input ceramic capacitor loop

2. Q1/Q2 switching loop

3. SW node size and shape

4. Bootstrap loop

5. DRVCC/INTVCC decoupling

6. Gate-driver loops

7. Rsense Kelvin routing

8. Feedback divider routing

9. U1 exposed-pad thermal copper

10. Half-bridge power loop

11. Output return current path

12. Separation between noisy SW/OUT nodes and feedback/control traces

The external engineering partner should not simply copy the current layout. The current board was useful as a prototype but requires a controlled revision.

Step 4: Simulate before fabrication

Required simulations:

1. 110VDC to 24V buck regulation

2. Startup into no load

3. Startup into load

4. 24V 100kHz half-bridge output

5. Buck response to half-bridge load

6. MOSFET VDS ringing

7. SW-node ringing

8. Output overshoot and TVS clamp current

9. Current-limit event

10. Thermal loss estimate

Simulation should include realistic MOSFET models and parasitic inductance estimates where possible.

Step 5: Prototype bring-up procedure

The next prototype must be brought up in stages.

Recommended bring-up:

1. No power:

- **Check all rails to ground.**
- **Check MOSFET gate-source resistance.**
- **Check no shorts on VIN, 24V_BUS, DRVCC, INTVCC, EXTVCC.**

2. Low-voltage buck test:

- **Apply 24V input.**
- **Force RUN high through 100kΩ if required.**
- **Keep half-bridge disabled.**
- **Confirm buck regulation.**

3. Intermediate input test:

- **Test at 48V, 80V, then 110V.**
- **Monitor input current, 24V_BUS, DRVCC, INTVCC, EXTVCC, SW, Q1/Q2 temperature.**

4. Half-bridge disabled thermal test:

- **Run for at least 10 minutes.**
- **Confirm U1, Q1, Q2, L1 temperatures.**

5. Enable half-bridge with no external load:

- Confirm 24V 100kHz waveform.

- Monitor EN, f_OUT, output waveform, 24V_BUS.

6. Load testing:

- Test with representative load.

- Increase load gradually.

- Monitor all MOSFET and controller temperatures.

7. Fault testing:

- Output open.

- Output short.

- Startup/shutdown cycling.

- Input voltage variation.

- Overcurrent trip.

Use a current-limited supply wherever possible. A fuse alone is not adequate protection during prototype bring-up.

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14. Acceptance Criteria

The external engineering partner should validate at least:

1. Stable 24V bus regulation.

2. Stable 24V 100kHz square-wave output.

3. No skipped output pulses under normal conditions.

4. Correct startup and shutdown behaviour.

5. No uncontrolled heating of LTC7897, Q1, Q2, Q5, Q6, L1, or Rsense.

6. Acceptable SW-node ringing and MOSFET VDS margin.

7. Output TVS does not conduct during normal operation.

8. Current protection trips correctly.

9. PGOOD/EN sequencing works correctly.

10. Board survives repeated power cycling.

Thermal pass/fail limits should be agreed before validation.

Suggested preliminary validation target:

No semiconductor case temperature above 80°C at maximum specified ambient and maximum specified load, unless the component datasheet and reliability analysis justify a higher value.

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15. Summary of Main Design Lessons From Current Prototype

The current prototype confirmed that the architecture can work, but it also identified several necessary changes.

Main lessons:

1. Q2 gate damping is mandatory.

2. The oscillator output resistor must be low value, not 220kΩ.

3. The LTC7897 driver supply voltage must be tunable.

4. The LTC7897 thermal layout must be improved.

5. The SW snubber must be tuned, not blindly fitted.

6. The half-bridge driver supply must be separated from the new 24V bus.

7. The output TVS and capacitors must be redesigned for 24V.

8. The next PCB must include proper test points and configuration options.

9. The heavily reworked current PCB should not be used for final validation.

The current heavily reworked PCB should not be used as the final validation reference. It should be treated as a proof-of-concept prototype that exposed the required revision items.

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16. Immediate Deliverables Required From External Engineering Partner

The external engineering partner should provide:

1. Revised schematic for 110VDC input to 24V 100kHz output.

2. Updated calculations:

- buck feedback divider

- inductor ripple

- current limit

- MOSFET losses

- gate-drive losses

- thermal estimates

3. Updated BOM with voltage/current/thermal margins.

4. Revised PCB layout.

5. Gerber files.

6. Assembly files.

7. Simulation files.

8. Bring-up procedure.

9. Test report template.

10. Final validation report after prototype testing.

No production release should occur until a revised prototype has passed electrical, thermal, and fault testing.